

Major Histocompatibility Complex and immunity

when a transportation happen
and the body either accept
or reject the organ

Course outline

1. Discovery
2. Nomenclature
3. The function
4. Genes of MHC
5. Protein structure
6. Ag processing and presentation
7. MHC bind self peptides
8. Other Ags that activate T cells

Discovery

How the MHC got its name

The term derives from research in transplantation that provide insight into the rules governing the acceptance or rejection of tissues (histocompatibility).

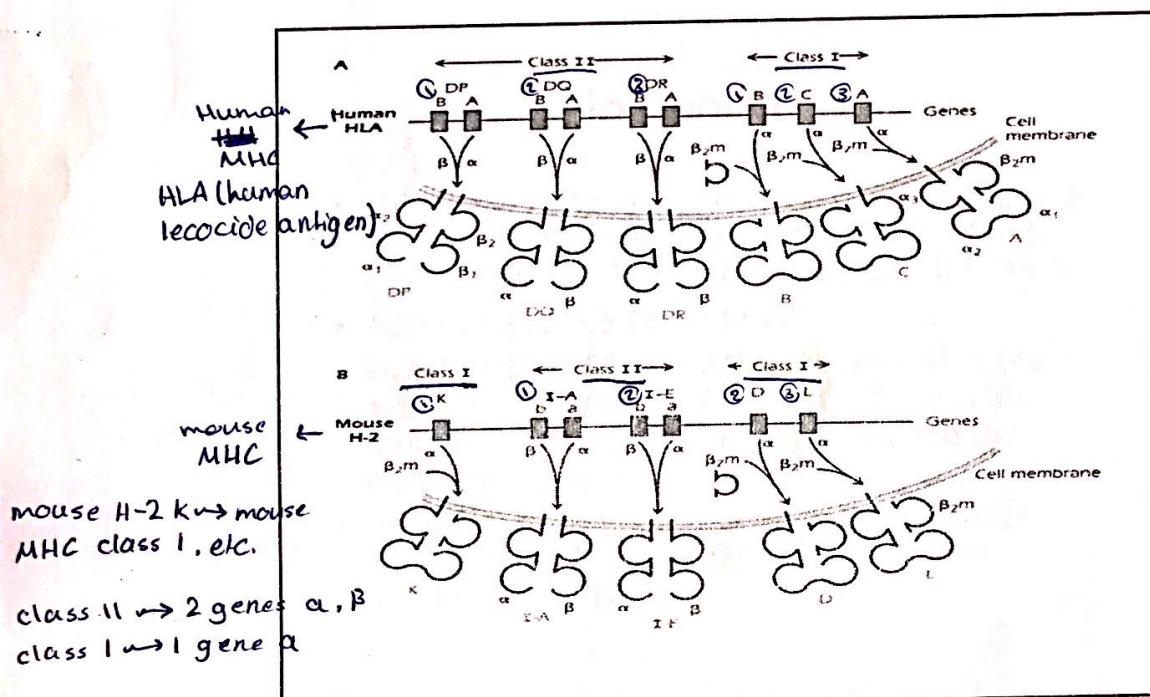
Interaction with T cells

- The function of MHC molecules are:

Ag presentation

to bind peptides that are produced when proteins are catabolized and to present these peptides to T cells

present on the membrane and present the antigen for T cell



Genes of MHC

- MHC genes are complex inherited as a unit. Includes two classes:
- **Class I region** contains 3 independent genes.
the mouse MHC is called H-2: K, D, L
the human MHC is called HLA: A, B, Cw
Another gene does not belong to MHC but class I molecules are associated with it is called β 2-microglobulin
- **Class II region** contains 2 or 3 independent genes:
Murine H-2: I-A and I-E
Human HLA: DP, DQ, DR

Nomenclature

- Originally, HLA molecules were characterized by serology using specific antibodies. So HLA-A2 differ from HLA-A25, HLA-DR14 from HLA-DR-16
- According to PCR technology
HLA-DRB1*13 01: the letters to the left of the asterisk specify the HLA gene (DRB1) and the numbers to the right (4 or 6 or 8) define the particular allele
- In mice, H-2 haplotype are given superscript letters e.g. H-2^b, in human, HLA are given numbers e.g. HLA-A4

MHC gene characteristics

- **Polygenic:** MHC class I and II molecules are coded by multiple independent genes. In human, 3 indep. Genes A, B, C codes for class I molecules expressed at the cell surface.
Each cell has 2 sets of chromosomes (1 maternal and 1 paternal)
Also 3 different MHC class II molecules DP, DQ, DR

- **Polymorphism**

multiple stable forms of each MHC genes exist in the population. MHC is the most highly polymorphic gene system in the population. *in the population*
Hundreds of slightly different versions (alleles) of each of the MHC genes are identified.

The enormous diversity of MHC genes is the major barrier to successful transplantation of organs.

Although MHC molecules are diverse within the population, each cell in a particular person (liver, kidney.....) expresses the same set of HLA class I and II molecules.

ex:
in humans, all humans have 3 different blood alleles that every human must have any of them, but there's no human that can have the three alleles together. same with MHC, human population have hundreds of alleles but it's impossible for any human to have more than 2 MHC alleles that are identical in the same individual for all cells but there are no two humans that have the same MHC

- **Codominant expression** (both alleles are expressed)
every cell expresses MHC molecules transcribed from both the maternal and paternal chromosomes. Thus, every nucleated cell of a particular person expresses up to 6 different HLA class I and 6 different HLA class II molecules. Some individuals express fewer than 6 because they are homozygous for a particular allele

- **Coordinate regulation** (all 6 genes exist on the same cell)
All class I and II molecules are expressed at the same time on a single cell

Class I and II are regulated separately: class I can be expressed in the absence of class II molecules

in one population there are 200 different classes
HLA class I: A-1 B-33 C-200

class II: DR-76 DP-22 DQ-70

that's why there are no two identical humans HLA

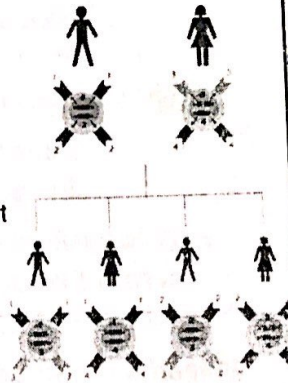
• Inheritance:

-A **haplotype** (the set of HLA class I and II genes expressed on an individual chromosome) is passed on to offspring as a unit

-Each parent expresses one haplotype different from each child and each child one haplotype different from each parent.

-Every child in the family expresses a different set of HLA haplotype (1 in 4 chance that 2 siblings will have identical haplotype)

-The Likelihood of 2 individuals in the general population who are matched at all HLA alleles is extremely small



* siblings are more HLA identical than ~~sub~~ children with parents because children take from every parent maximally 50% identical while in siblings it can be more

Other genes within HLA

In the class I region, HLA-E, HLA-F, and HLA-G

Function: unknown but HLA-G expressed by placental trophoblast cells prevent the rejection of the fetus by the maternal host. *the mother get an extra immune compromise (immunity become weaker)*

class II region includes HLA-DM involved in peptide exchange in the MHC class II pathway for Ag presentation, TAP-1 and 2 involved in the MHC class I Ag presentation pathway

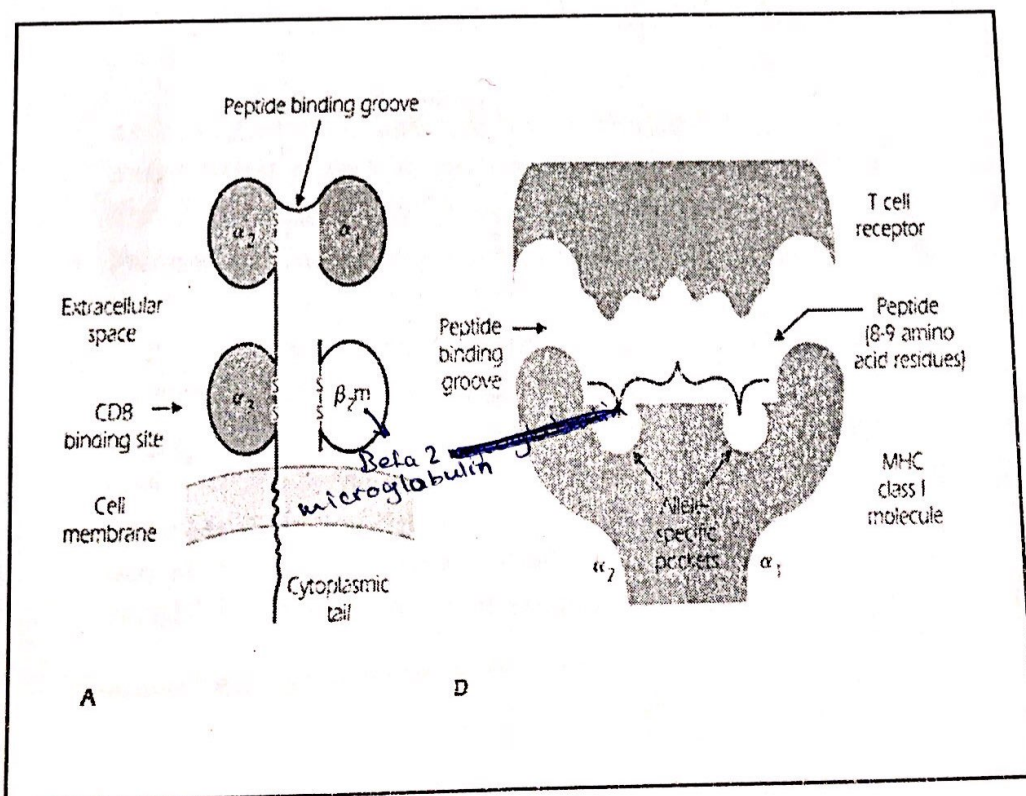
after the gene is expressed into a protein

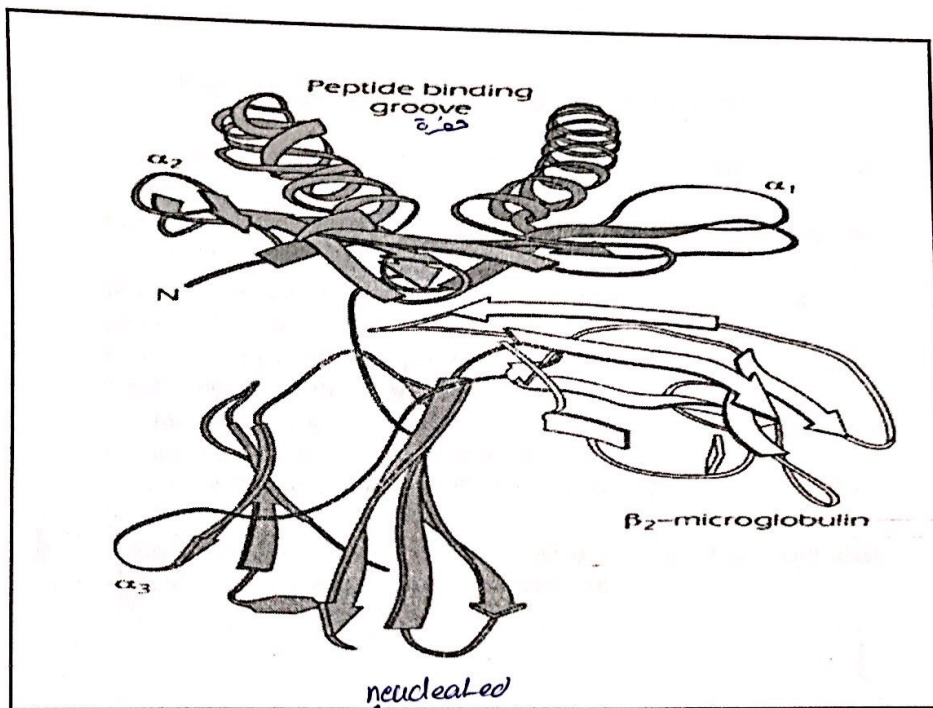
At the protein level Structure of MHC class I molecules

- A transmembrane glycoprotein expressed in noncovalent association with a small invariant (identical) polypeptide called β_2 -microglobulin.
- Consists of α heavy chain that comprises 3 extracellular Ig-like domains α_1 , α_2 and α_3 .
- β_2 -microglobulin has a structure homologous to a single Ig domain. The α_1 is paired with α_2 and α_3 with β_2 -microglobulin. that stabilize the chain
- Has a peptide-binding groove that resemble a basket with an irregular floor (β -pleated sheet) surrounded by walls (α -helices). The groove holds 8-9 a.a. long

β_2m only exist in class I

~~do not exist on major histocompatibility complex~~

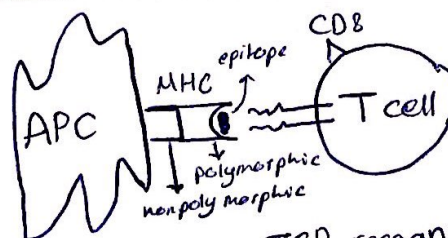




class I is present on all nucleated cells and are identical
 red blood cells at first has erythroblast then becomes erythrocyte without MHC nor a nucleus

- An individual MHC molecule can be divided into a polymorphic or variable region and a nonpolymorphic region that is similar in all MHC class I molecules.
- **Presence:** found on all nucleated cells of an individual and are identical
- CD8 binds to the nonpolymorphic region of all MHC class I molecules, specifically in the α₃ domain
- Binding of an MHC class I molecule to a peptide is selective. One MHC molecule will bind with high affinity to only certain peptides. A single MHC molecule can bind to a variety of peptides but binds preferentially to peptides with certain motifs (closely related a.a. at certain positions).

several epitopes bind one MHC

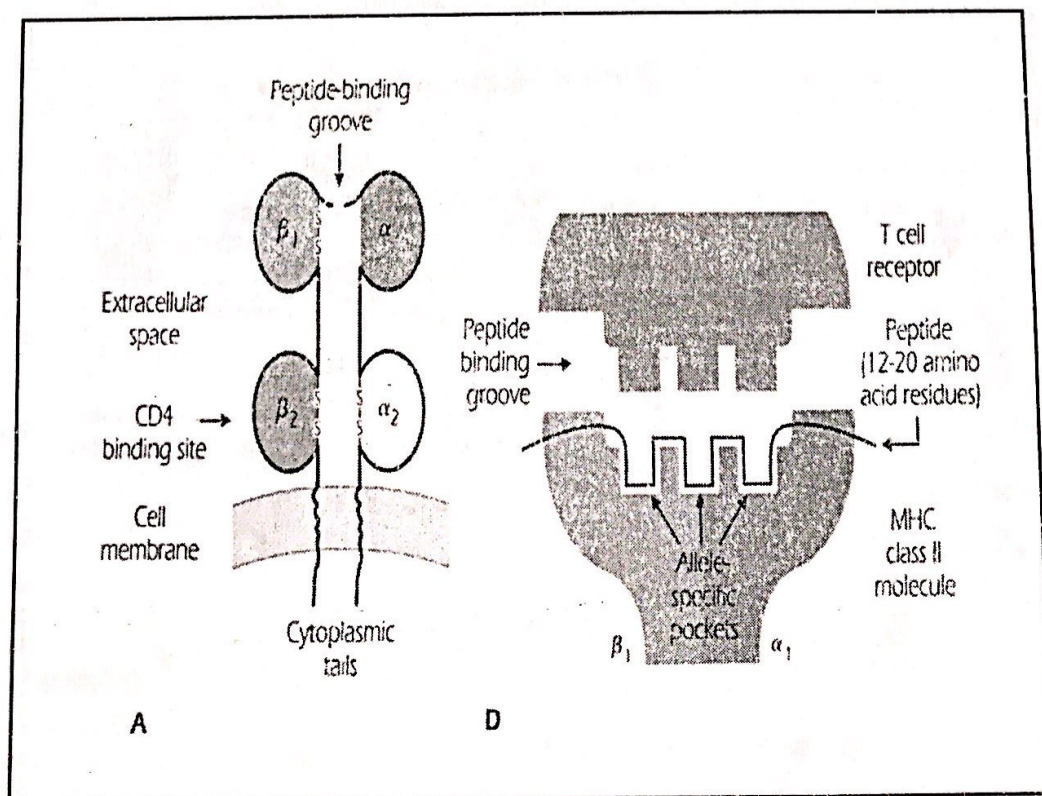


TCR recognize MHC and the epitope

Structure of MHC class II molecules

- A transmembrane glycoprotein comprising 2 chains α and β . Like class I molecules.
- Every MHC molecule is expressed at the cell surface with 4 domains. The α_1 is paired with β_1 and α_2 is paired with β_2 .
- Like class I molecules: class II molecule contains a peptide-binding groove at the top of the molecule.
- In contrast to the 8-9 a.a peptide that bind class I, the MHC molecule binds peptides between 12-17 a.a
- Like MHC class I molecules, MHC class II molecules are composed of polymorphic and nonpolymorphic regions. CD4 T cells binds to the nonpolymorphic region of all MHC class II molecules, specifically in the β_2 domain.
- Presence: on some immunological cells including APCs, thymic epithelial cells, Mac in the mice, activated T cells in human

class one is present only on some immune cells like APC (dendritic cells) and B cells



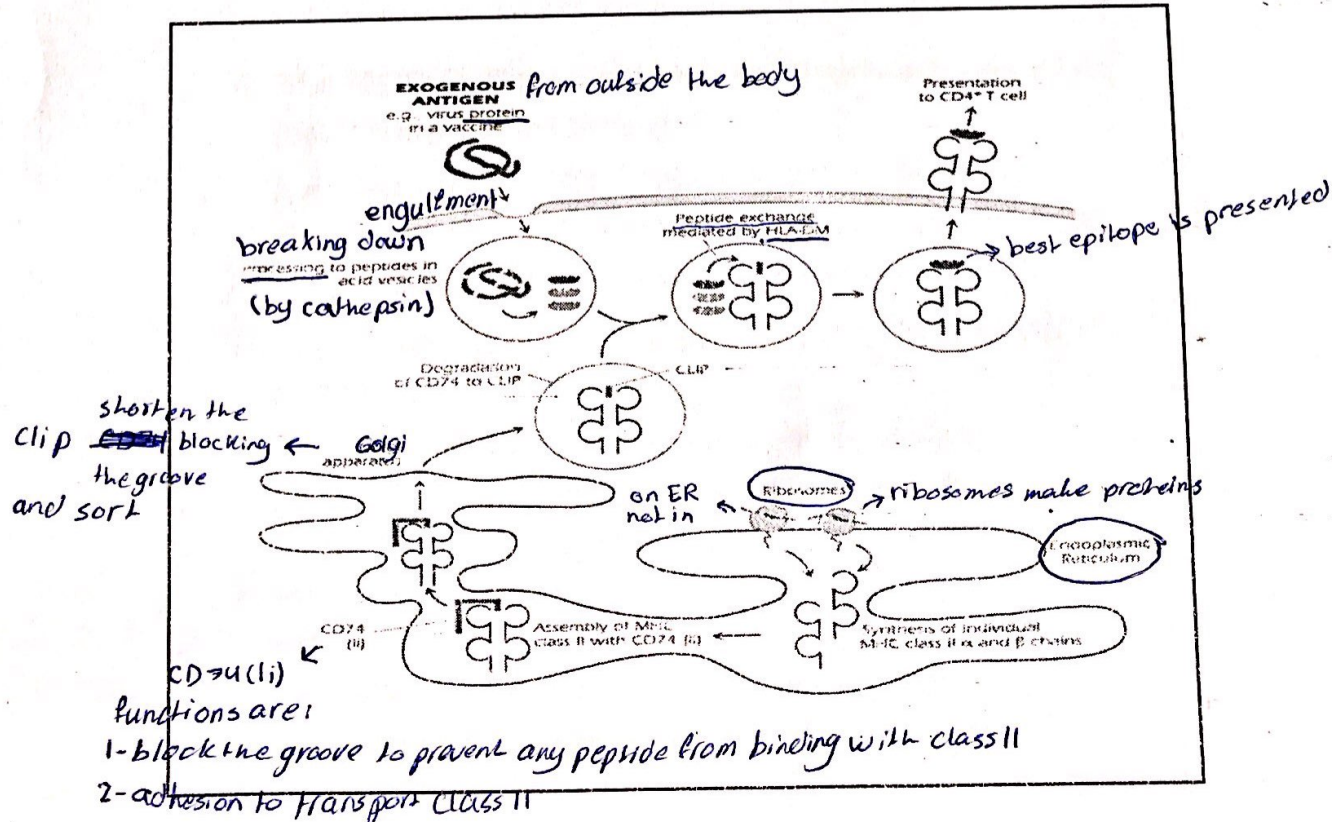
Ag Processing and presentation

Exogenous Ags: are Ags that come from outside a host cell and are taken inside by endocytosis or phagocytosis. Ags derived from bac, viruses or vaccines that do not injure the host.

Ag presenting cells (APC): dendritic cells, mac, and B cells take up exogenous Ags

- **Ag presentation**

1. Endocytosis: the protein is internalized in an intracellular vesicle that are highly acidic (pH4). These contains proteases as cathepsins that cut proteins into peptides to yield several peptides



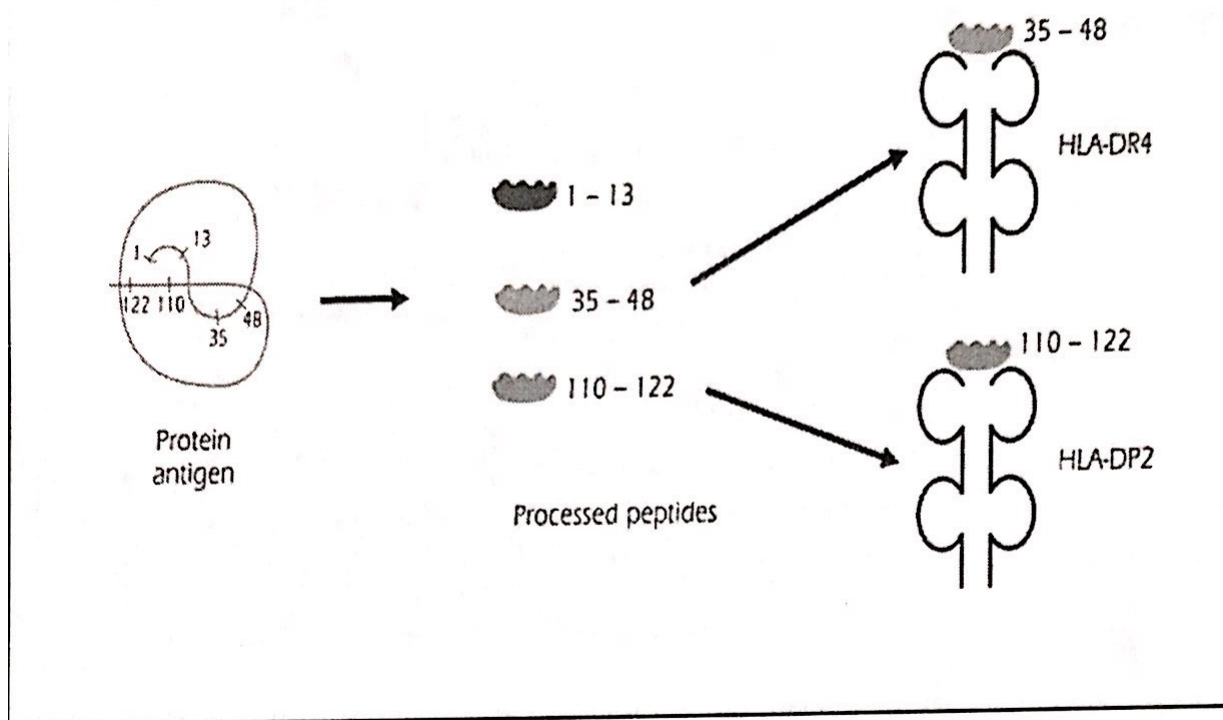
2. The MHC class II are synthesized on ribosomes and are assembled in the ER with an invariant chain (Ii).
 - a. Ii binds the groove of MHC preventing the binding of other peptides.
 - b. Ii act also as a chaperone for the newly synthesized MHC. This allows MHC to leave the ER and enters the Golgi.
3. In Golgi, removal of Ii occurs in stages. Ii is partially degraded leaving a fragment known as Class II-associated Invariant polypeptide (CLIP)

4. The MHC-CLIP leaves Golgi and enters acid vesicles (endosomes or lysosomes).

A molecule HLA-DM catalyze peptide exchange between the MHC-CLIP and peptides derived from exogenous Ags to form MHC-peptides.

5. This complex moves to the cell surface to present the peptide.

Although catabolism of a protein yields several peptides, not all the peptides bind to MHC because binding is selective. Unbound Peptides to MHC, do not trigger CD4+ T cell response. Every person will make a T cell response, because of selectivity, individuals who express different MHC molecules respond to different parts of the same protein

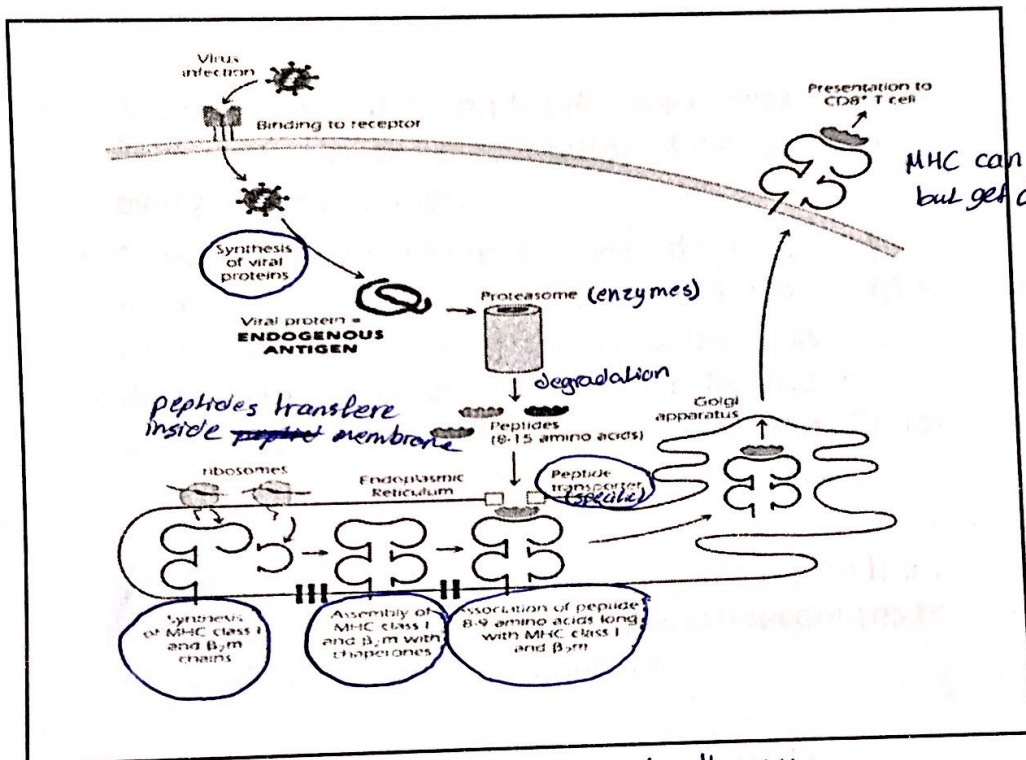


virus / tumor Endogenous Ags

- These Ags are synthesized inside a cell and are derived from pathogens that have infected the host cell.
1. Processing occurs in the cytoplasm via a giant E known as the Proteasome that cleaves proteins into peptides about 15 a.a. in length.
 2. Some peptides are destroyed but some are transported into ER by a peptide transporter (a product of 2 genes TAP-1 and TAP-2)

* exogenous $\rightarrow$ CD4
 endogenous $\rightarrow$ CD8

11/24/2018



* in general 95% of T cells ^{that enter thymus} die either because it failed to recognize epitopes or it recognized self peptides

3. MHC class I and β 2-microglobulin are synthesized separately and associate in the ER. The chaperon (tapasin) direct the transport of the complex to Golgi. MHC class 1 molecules preferentially bind peptides 8-9 a.a in length.

4. The MHC class I-peptide complex is presented on the surface to CD8+ T cell

- Because MHC class I molecules are expressed on all nucleated cells, the presentation of Ags occur in every cell in the body.
- Some viruses inhibit the synthesis of MHC class I molecules or interrupt the transport of peptide-MHC complexes to the cell surface. Thus, the virus decreases expression of MHC molecules and so decrease the CD8 T cell response to the virus. Tumor cells show decreased expression of MHC class I compared to normal cells
- Although the decrease in MHC class I decrease the T cell response, it may trigger the NK cell responses to the viral-infected cell or tumor cell.

*immune system treat viruses and tumors the same way
 *viruses and tumor cells lose MHC class I (escape mechanism)
 because MHC is important for binding with T cells, when it's absent the immune system will fail to recognise the cell

• Cross-presentation

Exogenous Ags are presented in the MHC class I pathway.

- Proteins from bac, most viruses, allergens and harmless Ags all trigger CD4+ T cell. Infectious pathogens activate class I associate with MHC molecules pathway
- The processing pathway rather than inherent properties of the Ag, determine whether a protein is presented to CD4 or CD8 T cells.
- Self proteins bind MHC class I and class II molecules but do not normally activate T cells.

Other Ags that activate T cell response

1. Lipids and Glycolipids

- Some T cells recognize lipids and glycolipids found in the cell wall of pathogens such as *M. tuberculosis* that live inside Mac.
- The lipids are presented by a family of molecules known as CD1 expressed by APC.
- CD1 are surface glucoprotein coded for outside the MHC region. Like MHC class I molecules, they are expressed on the surface of APC in association with $\beta 2$ -m. The structure also similar to MHC class I with a larger groove. The deep groove binds the hydrophobic part of the lipid Ag, and the polar region binds the TCR.
- HSV 1 downregulates expression of CD1 and thus inhibit Ag presentation
- CD1a,b,c present lipids to T cells with TCR, but CD1d present lipids to NKT cells.

• SuperAg:

-Predominately bac exotoxins such as S enterotoxin.

-is not processed but the intact molecule binds to MHC class II molecules outside the groove.

SuperAgs activate up to 10% of total T cells.

This massive activation release CK that is responsible for toxic shock

characteristics!

1- doesn't bind the group

2- no processing (no cleavage)

3- bind β regardless of α

4- many T cells are specific for superantigen

5- 10% of T cells are activated because α is non-self (massive activation of cytokines that are responsible for toxic shock)

